EXP-6 Retrieval of Similar Items using Vector Embeddings (FAISS)

AIM:

To implement a system for retrieving similar movies using high dimensional vector embeddings with FAISS.

DATASET:

MovieLens Dataset (Movie titles)

PROCEDURE:

Step 1: Install and Import Libraries pip install faiss-cpu sentence-transformers pandas

import faiss import pandas as pd from sentence_transformers import SentenceTransformer

Step 2: Load Dataset

```
movies = [
"The Matrix",
"Inception",
"Interstellar",
"The Dark Knight",
"Titanic",
"Avatar",
"The Avengers"
]
df = pd.DataFrame(movies, columns=["title"]) print(df.head())
```

Output:

```
title
0        The Matrix
```

1 Inception

2 Interstellar

3 The Dark Knight

4 Titanic

Step 3: Load Embedding Model

model = SentenceTransformer('all-MiniLM-L6-v2')

Step 4: Convert Text to Embeddings

embeddings = model.encode(df['title'].tolist()) print(embeddings.shape)

Output:

(7, 384)

Step 5: Create FAISS Index

dimension = embeddings.shape[1] index = faiss.IndexFlatL2(dimension) index.add(embeddings)

Step 6: Perform Similarity Search

query = "science fiction movie" query_vec = model.encode([query]) distances, indices = index.search(query_vec, k=3)

results = df.iloc[indices[0]] print(results)

Output:

title

0 The Matrix

2 Interstellar

1 Inception

Step 7: Observations

- FAISS retrieves semantically similar movies
- Embeddings capture meaning, not just keywords
- Results are relevant to query context

Output:

Top 3 similar movies retrieved successfully.

RESULT:

FAISS retrieves similar movies efficiently using high-dimensional embeddings. It enables accurate semantic search for recommendation systems.